

Week 1 Module – Programming Practice (3rd Year)

Date: 10/06/2026

Language used: JAVA

Pdf follows: handwritten code with logic and followed by the respective outputs.

Submission details:---

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Week 1 module - Programming Practice

Technical Training - 10/06/2026.

Concept: Pattern.

* Pattern uses nested loops concept.
* For loop → best used when iterations are known. more compact
* While loop → " " " " " " not known, more flexible.

Programming

CODE: - A) Star problem pattern.
B) Number pattern problem.

```
import java.util.Scanner;  
public class PatternA {  
    public static void main (String [] args) {  
        Scanner sc = new Scanner (System.in);  
        System.out.println ("Enter no. of lines n: ");  
        int n = sc.nextInt(); // takes user input
```

// Star Pattern.

/* Question 1: - samare

```
*****  
*****  
*****  
*****  
*****
```

... comment out to present the Q*/.

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= n; j++) {  
        System.out.print ("* "); // prints *  
    }  
    System.out.println(); // moves to next line.
```

}

- Logic -

i = 1 → i <= 5 is true. used to separate Q1 from Q2.

n = 5
↓
j = 1 → j <= 5 true → prints ***** in row 1
again loops till cond == false. → moves to next line → print line

Output of pattern 1:

```
PS C:\Users\saksh\OneDrive\Desktop\coding\JAVA> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\saksh\AppData\Roaming\Code\User\workspaceStorage\00d06a9f6bea38ef6bd2abb66d2488b9\redhat.java\jdt_ws\JAVA_c8f87ea5\bin' 'TechnicalTraining.week1.pattern_problems.A'  
Enter number of lines n:  
5  
*****  
*****  
*****  
*****  
*****
```

n=5

/* Question 2: Hollow Square.

clearly,

i=1 → 5 stars

i=5 → 5 stars

i.e. i=1 || i=n

similarly,

j=1 || j=n

→ 5 stars

∴ if-else cond.
is used.

prints space when
if condition is not
satisfied.

LOGIC

→ * * * * *

* * * * *

* * * * *

* * * * *

→ * * * * *

for (int i=1; i<=n; i++) {

for (int j=1; j<=n; j++) {

if (i==1 || i==n || j==1 || j==n) {

system.out.print(" * ");

} else {

system.out.print(" ");

}

system.out.println(); // used to separate

Q2 from Q3.

/* Question 3: Triangle.

LOGIC

n=5

i=1 → i<=5 ✓

j=1

j<=i ✓

→ *

moves next line

i=2 → i<=5 ✓

j=1

j<=i ✓

→ *

next line

*

* *

* * *

* * * *

* * * * *

for (int i=1; i<=n; i++) {

for (int j=1; j<=i; j++) {

system.out.print(" * ");

system.out.println();

system.out.println();

system.out.println();

system.out.println();

* *
* *
* *

*
**

n-i+1... why? **Question 4: inverted triangle.**

⇒ Because

Row 1 → 5 *
2 → 4 *
3 → 3 *
4 → 2 *
5 → 1 *

if n=5.
clearly → j should follow (n-i+1)
i.e. $n-i+1$
 $5-1+1=5$
prints 5 stars respectively.

```
*****
****
***
**
*
```

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= (n-i+1); j++) {
        system.out.print(" * ");
    }
}
```

```
    system.out.println();
```

```
    system.out.println(); // separates Q4 from Q5
```

Question 5: Pyramid.

⇒ - represents space.

Logic → space stars space

```

  - - - * - - -
 - - * * * * - -
- - * * * * * -
 * * * * *

```

using 3 loops.
i = rows
j = columns → space
k = columns → stars

d ⇒ $n-i+1$ $2*i+1$ $n-i+1$

```
for (int i = 0; i < n; i++) {
    for (int j = 0; j < n-i+1; j++) {
```

```
        //space
        system.out.print(" ");
```

```
    }
    for (int k = 0; k < (2*i+1); k++) {
```

```
        //star
        system.out.print(" * ");
```

```
    }
    system.out.println();
```

```
    }
    // separating Q5 to Q6.
```

```
*****
****
***
**
*
```

```

  *
 ***
*****
*****
*****

```


/* Question 6 :

i = 0;
i < n.

```

      *
     **
    ***
   ****
  *****
 *****
  *****
   ****
    ***
     **
      *
  
```

i = n-1
i starts with
n-1 and decrease.

j < n-i-1
k < (2*i+1)

↑ upper half
↓ lower half.

j < n-i
k < 2*i-1.

// upper half

```

for (int i = 0; i < n; i++) {
    for (int j = 0; j <= n-i-1; j++) {
        System.out.print(" ");
    }

```

```

    for (int k = 0; k < (2*i+1); k++) {
        System.out.print("* ");
    }

```

```

    System.out.println();

```

```

    System.out.println(); // separates from
                          // Q6 to Q7.

```

// lower half

```

for (int i = n-1; i >= 1; i--) {
    for (int j = 1; j <= n-i; j++) {
        System.out.print(" ");
    }

```

```

    for (int k = 0; k < 2*i-1; k++) {
        System.out.print("* ");
    }

```

```

    System.out.println();

```

```

    System.out.println(); // separates Q6 to
                          // Q7.

```

```

*
***
*****
*****
*****
*****
***
*
  
```

/* Question 7 : Half Diamond.

```

*
**
***
****
*****
****
***
**
*

```

upper half
 $j \leq i$

lower half

$j \leq i$

here
i starts with
 $i = n - 1$

condition changes

```

for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print("*");
    }
    System.out.println();
}
for (int i = n - 1; i >= 1; i--) {
    System.out.print("*");
    for (int j = 1; j <= i; j++) {
        System.out.print("*");
    }
    System.out.println();
}

```

System.out.println(); // separates 97 to 98.

/* Question 8 : Half Diamond inverted.

```

*****
****
***
**
*

```

```

for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        System.out.print("*");
    }
    System.out.println();
}

```

System.out.println();

Scanner.close(); // closing of scanner class, so obj.
// closing of main method.

```
*
**
***
****
*****
*****
****
***
**
*
```

output of pattern 7

Output of pattern 8 :

```
*****
****
***
**
*
PS C:\Users\saksh\OneDrive\Desktop\coding\JAVA> & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\saksh\AppData\Roaming\Code\User\workspaceStorage\00d06a9febea38ef6bd2abb66d2488b9\redhat.java'
\jdt_ws\JAVA_c8f87ea5\bin' 'TechnicalTraining.week1.pattern_problems.A'
```

Ln 104, Col 14 Spaces: 4 UTF-8 CRLF Java Go Live Prettier

```
import java.util.Scanner;
public class B {
    public static void main (String[] args) {
        Scanner sc = new Scanner (System.in);
        System.out.println ("Enter no. of lines n: ");
        int n = sc.nextInt();
```

//number Pattern

/* Question 1: *number square*.

```
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1
1 1 1 1 ... *1
```

uses same
logic like

↓ square pattern.

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        System.out.print ("1");
    }
    System.out.println ();
}
```

System.out.println (); // separates Q1 from Q2.

/* Question 2: *incrementing*.

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5 ... *1
```

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        System.out.print (j + " ");
    }
    System.out.println ();
}
```

System.out.println (); // separates Q2 to Q3


```
PS C:\Users\saksh\OneDrive\Desktop\coding\JAVA> c::; cd 'c:\Users\saksh\OneDrive\Desktop\coding\JAVA'; & 'C:\Program Files\Java\jdk-24\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\saksh\AppData\Roaming\Code\User\workspaceStorage\00d06a9febea38ef6bd2abb66d2488b9\redhat.java\jdt_ws\JAVA_c8f87ea5\bin' 'TechnicalTraining.week1.pattern_problems.B'  
Enter number of lines n:  
5  
1 1 1 1 1  
1 1 1 1 1  
1 1 1 1 1  
1 1 1 1 1  
1 1 1 1 1
```

Output of number pattern 1

```
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5  
1 2 3 4 5
```

Output of number pattern 2

/* question 3: right triangle:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5 ... */
```

```
for (int i=1; i<=n; i++) {
    for (int j=1; j<=i; j++) {
        system.out.print(j + " ");
    }
    system.out.println();
}
```

system.out.println(); //separates Q3 to Q4

/* question 4: inverted diamond.

```
3 3 3 3
4 4 4 4
5 5 5
6 6
7 ... */
```

```
for (int i=n; i>=1; i--) {
```

to start with → int num = 2+i;
3 ⇒ 2=1
num ⇒ 2+1=3
for (int j=1; j<=i; j++) {
 system.out.print(num + " ");

and afterwards ↓
num++;
system.out.println();

system.out.println(); //separates Q4 to Q5

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

1 * Question 9 : sandwich pattern.

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3 ...

$i < 0$
 $i < n1 \Rightarrow i < 3$
loop runs

upper
half.

```
int n1 = 3;
for (int i = 0; i < n1; i++) {
    for (int s = 0; s < i; s++) {
        system.out.print(" ");
    }
    int count = 5 - (2 * i);
    for (int j = 0; j < count; j++) {
        system.out.print((3 + i) + " ");
    }
    system.out.println();
}
```

lower
half

```
for (int i = n1 - 2; i >= 0; i--) {
    for (int s = 0; s < i; s++) {
        system.out.print(" ");
    }
    int count = 5 - (2 * i);
    for (int j = 0; j < count; j++) {
        system.out.print((3 + i) + " ");
    }
    system.out.println();
}
```

```
system.out.println(); // separates Q5 & Q6
```

```
3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3
```

10 Question 6: incremented half diamond.

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3

```

↑ upper
↓ lower.

```

3 // ... */
for (int i=0; i<n; i++) {
    int num = i+3;
    for (int j=0; j<=i; j++) {
        system.out.print(num + " ");
    }
    system.out.println();
}

for (int i=n-2; i>=0; i--) {
    int num = i+3;
    for (int j=0; j<=i; j++) {
        system.out.print(num + " ");
    }
    system.out.println();
}

sc.close(); //closing of scanner obj.
//closing of main func.
} //closing of public class.

```

```

3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
PS C:\Users\saksh\OneDrive\Desktop\coding\JAVA>

```


